

KYIV SCHOOL OF ECONOMICS

Department of Economics · EBD and BE

PROBABILITY & STATISTICS MOCK MIDTERM EXAMINATION

This test is not for grading!

Summer Term, 2026 | Instructors: Rozora I., Miroshnichenko I.,
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Full Name: _____ **Score:** _____/100
Group: _____ **Date:** _____

Part	Section	Format	Items	Points
I	A Multiple Choice	circle one	10	10
	B True or False	T/F + correct	8	8
	C Validity Check	show work	2 PMF + 2 PDF	6
	D R Output (4 blocks)	interpret	16 questions	16
<i>Part I subtotal</i>				40
II	Problem 1	show work	sub-parts	6
	Problem 2	show work	sub-parts	6
	Problem 3	show work	sub-parts	6
	Problem 4	show work	sub-parts	6
	Problem 5	show work	sub-parts	6
	Problem 6	show work	sub-parts	6
	Problem 7	show work	sub-parts	6
	Problem 8	show work	sub-parts	6
	Problem 9	show work	sub-parts	6
	Problem 10	show work	sub-parts	6
<i>Part II subtotal</i>				60
TOTAL				100

Instructions: (1) Show all work in Part II unjustified answers receive zero credit. (2) Standard Normal Table and Distributions Summary on the last pages. (3) No electronic devices or notes. (4) Write legibly.

	A	B	C	D	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	Total
Max	10	8	6	16	6	6	6	6	6	6	6	6	6	6	100
Score															

PART I: THEORY**(40 points)****SECTION A** Multiple Choice

[10 pts]

Circle the **one** correct answer. 10 questions, **1 point each**. No partial credit.

1. Which correctly characterises a **Probability Mass Function** (PMF)?

- (A) $\int_{-\infty}^{\infty} p(x) dx = 1$ and $p(x) \geq 0$.
- (B) $p(x) \geq 0$ for all x , and $\sum_{\text{all } x} p(x) = 1$.
- (C) $p(x)$ must be a continuous function of x .
- (D) $p(x)$ may exceed 1 for individual values of x .

2. Let X have $E[X] = 3$ and $\text{Var}(X) = 4$. What is $\text{Var}(2X + 5)$?

- (A) 8 (B) 13 (C) 16 (D) 21

3. If $Z \sim N(0, 1)$, which statement is **FALSE**?

- (A) $P(Z > 0) = 0.5$.
- (B) $P(Z = 0) = 0.5$.
- (C) $E[Z] = 0$ and $\text{Var}(Z) = 1$.
- (D) The distribution is symmetric about 0.

4. Markov's inequality: for a **non-negative** r.v. X with $E[X] = \mu$ and any $a > 0$:

- (A) $P(X \geq a) \leq \frac{\mu}{a}$.
- (B) $P(X \leq a) \leq \frac{\mu}{a}$.
- (C) $P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$.
- (D) $P(X \geq a) = e^{-a/\mu}$.

5. $X \sim \text{Binomial}(n = 20, p = 0.30)$. What is $\text{Var}(X)$?

- (A) 0.21 (B) 0.30 (C) 4.2 (D) 6

6. The **memoryless property** of the Exponential distribution states, for all $s, t > 0$:

- (A) $P(X > s + t \mid X > s) = P(X > t)$.
- (B) $E[X^2] = (E[X])^2$.

(C) $\text{Var}(X) = E[X]$.

(D) $P(X > s + t) = P(X > s) + P(X > t)$.

7. The CLT states that $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$ converges in distribution to:

- (A) $N(\mu, \sigma^2)$ (B) $N(0, 1)$ (C) $\text{Uniform}(0, 1)$ (D) $\text{Exponential}(1)$

8. Which value indicates the **strongest** linear relationship?

- (A) $\rho = 0.72$ (B) $\rho = -0.91$ (C) $\rho = 0.50$ (D) $\rho = 0.01$

9. A credit-risk model flags applicants. 5% of the population is truly high-risk. The model detects 90% of high-risk applicants and falsely flags 10% of low-risk ones. Given a positive flag, the probability the applicant is *actually* high-risk is closest to:

- (A) 0.05 (B) 0.32 (C) 0.50 (D) 0.90

10. The **Law of Large Numbers** guarantees that, as $n \rightarrow \infty$:

- (A) $\bar{X}_n$ converges in probability to the population mean μ .
(B) $\bar{X}_n$ converges in distribution to $N(0, 1)$.
(C) Individual observations X_i become closer to μ .
(D) The sample variance converges to 0.

SECTION B True or False

[8 pts]

Mark **T** (True) or **F** (False). If **FALSE**, write the corrected statement in the right column. 8 statements, 1 point each.

#	Statement	T / F	Correction if False
1	For any events A and B : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.		
2	If $\text{Cov}(X, Y) = 0$, then X and Y are independent.		
3	For any <i>continuous</i> r.v. X and constant c : $P(X = c) = 0$.		
4	The variance of a random variable can be negative if the distribution is left-skewed.		
5	The Law of Large Numbers guarantees $\bar{X}_n \xrightarrow{p} \mu$ as $n \rightarrow \infty$.		
6	If $X \sim N(\mu, \sigma^2)$, then $\frac{X - \mu}{\sigma} \sim N(1, 1)$.		
7	Markov's inequality: for $X \geq 0$ and $a > 0$, $P(X \geq a) \leq \frac{E[X]}{a}$.		
8	The correlation satisfies $-1 \leq \rho \leq 1$, and $\rho = \pm 1$ iff there is a perfect <i>linear</i> relationship.		

SECTION C Validity Check

[6 pts]

Verify **both** conditions: (i) non-negativity, (ii) sum or integral equals 1. Show all work. State your conclusion clearly. **1.5 pts each.**

C1. [1.5 pts] Is the following a valid **PMF**?

$$f(x) = \frac{x+1}{10}, \quad x \in \{0, 1, 2, 3\}, \quad f(x) = 0 \text{ otherwise.}$$

Valid PMF? _____

Work:

C2. [1.5 pts] Is the following a valid **PMF**?

$$f(x) = \frac{x+1}{8}, \quad x \in \{1, 2, 3\}, \quad f(x) = 0 \text{ otherwise.}$$

Valid PMF? _____

Work:

C3. [1.5 pts] Is the following a valid **PDF**?

$$f(x) = 3x^2, \quad 0 \leq x \leq 1, \quad f(x) = 0 \text{ otherwise.}$$

Valid PDF? _____

Work:

C4. [1.5 pts] Is the following a valid **PDF**?

$$f(x) = 2x, \quad 0 \leq x \leq 2, \quad f(x) = 0 \text{ otherwise.}$$

Valid PDF? _____

Work:

C3. [+2 bonus pts] Find $k > 0$ such that $f(x) = k \cdot x(4 - x)$, $x \in \{1, 2, 3\}$, $f(x) = 0$ otherwise, is a valid **PMF**. Verify both conditions for your k .

SECTION D R Output Interpretation

[16 pts]

Four independent blocks, **4 pts each** (1 pt per question). Answer in complete sentences. **Interpretation only: no arithmetic required!**

Block D1 - Normal Distribution [4 pts]

A music analyst models daily stream counts (thousands) for Ukrainian pop songs as $X \sim N(\mu = 12, \sigma = 3.2)$.

```
> pnorm(15.2, mean = 12, sd = 3.2)
[1] 0.8413
> pnorm(8.8, mean = 12, sd = 3.2)
[1] 0.1587
> pnorm(5.6, mean = 12, sd = 3.2, lower.tail = FALSE)
[1] 0.9772
> qnorm(0.90, mean = 12, sd = 3.2)
[1] 16.10
```

D1(a) `pnorm(15.2, ...)` = 0.8413 What fraction of songs receive *fewer than* 15,200 streams per day? Interpret for a music producer.

D1(b) The two lower-tail outputs give $P(8,800 < X < 15,200) = 0.8413 - 0.1587 = 0.6826$. Which well-known rule from lecture does this correspond to? Express $[8,800; 15,200]$ in terms of μ and σ .

D1(c) `pnorm(5.6, ..., lower.tail=FALSE)` = 0.9772 Rewrite as $P(X \dots) = \dots$ and interpret for an analyst tracking *underperforming* songs.

D1(d) `qnorm(0.90, ...)` = 16.10 A label considers a song a “hit” if it lands in the top 10% of daily stream counts. What is the hit threshold, and what does 16,100 represent geometrically on the normal curve?

Block D2 - Binomial Distribution [4 pts]

The campus coffee shop has a barista who makes an incorrect order with probability $p = 0.12$, independently each time. Let $X \sim \text{Binomial}(10, 0.12)$.

```
> dbinom(0, size = 10, prob = 0.12)
[1] 0.2785
> pbinom(1, size = 10, prob = 0.12)
[1] 0.6583
> 1 - pbinom(2, size = 10, prob = 0.12)
[1] 0.1087
```

D2(a) `dbinom(0,...)` = 0.2785 Interpret in plain language. Is this the probability of a “good” shift?

D2(b) `pbinom(1,...)` = 0.6583 What event does this probability describe? Is $X = 0$ or $X = 1$ included?

D2(c) `1 - pbinom(2,...)` = 0.1087 The manager intervenes when *more than 2* orders are wrong. What does this output represent? Interpret for the shop manager.

D2(d) Both `pbinom` commands compute cumulative probabilities. In your own words, what is the difference between `pbinom(1,...)` and `1 - pbinom(2,...)`? Why are their values so different?

Block D3 - Exponential Distribution [4 pts]

Late-night Bolt Food delivery times in Kyiv follow $T \sim \text{Exponential}(\lambda = 1/20)$, i.e. with mean $E[T] = 20$ minutes.

```
> pexp(30, rate = 1/20, lower.tail = FALSE)
[1] 0.2231
> pexp(40, rate = 1/20) - pexp(10, rate = 1/20)
[1] 0.4712
> qexp(0.50, rate = 1/20)
[1] 13.86
> qexp(0.95, rate = 1/20)
[1] 59.91
```

D3(a) `pexp(30,..., lower.tail=FALSE) = 0.2231` Interpret for a hungry student. What fraction of deliveries take *longer than* 30 minutes?

D3(b) `pexp(40,...) - pexp(10,...) = 0.4712` What probability does this expression compute? Describe the event in words.

D3(c) `qexp(0.50,...) = 13.86` This is the *median* delivery time. The mean is 20 minutes. Why is the median *less than* the mean for an exponential distribution? What does this imply about the shape?

D3(d) `qexp(0.95,...) = 59.91` The app promises: “95% of deliveries arrive within X minutes.” What is X ? Interpret for a customer deciding whether to order at 2 am.

Block D4 - Descriptive Statistics + Normal [4 pts]

A researcher records annual salaries (\$) of $n = 200$ employees at a Kyiv tech company.

```
> summary(salary)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2100   3800   4900   5600   6200  42000
> sd(salary)
```

```
[1] 3200

> # Assuming salary ~ N(mean = 5600, sd = 3200):
> pnorm(3800, mean = 5600, sd = 3200)
[1] 0.2869
> pnorm(9000, mean = 5600, sd = 3200, lower.tail = FALSE)
[1] 0.1440
```

D4(a) The mean (\$5,600) is notably higher than the median (\$4,900), and the maximum is \$42,000. What does this suggest about the *shape* of the salary distribution? Is the normal distribution likely a good fit here? Explain briefly.

D4(b) `pnorm(3800, ...)` = 0.2869 Interpret: what fraction of employees earn less than \$3,800/year under the normal approximation?

D4(c) `pnorm(9000, ..., lower.tail=FALSE)` = 0.1440 Interpret for an HR analyst. What fraction of employees are top earners (above \$9,000)?

D4(d) A manager states: “*Most of our employees earn above the average salary.*” Using the summary output (not the normal approximation), evaluate this claim. Is the statement true? What statistic is relevant here, and why?

PART II: APPLIED PROBLEMS**(60 points, 6 pts each)**

Show all work. Unjustified answers receive zero credit. Reference pages at the end.

Your university email server uses an AI spam filter. Historical data show:

- **20%** of all incoming emails are spam.
- The filter correctly detects **95%** of actual spam emails and flags them.
- The filter incorrectly flags **8%** of legitimate (non-spam) emails.

You open your spam folder and pick a random flagged email.

- Using the **Law of Total Probability**, find $P(\text{flagged})$.
- Apply **Bayes' theorem** to find $P(\text{spam} \mid \text{flagged})$.
- Your roommate says: “If the filter caught it, it’s definitely spam.” Evaluate this claim using part (b).
- In one day you receive 50 emails. How many do you expect to be spam, and how many of those does the filter correctly flag on average?

A KSE student “invests” their stipend in the DOGECAT meme token. Next month’s return R is modelled as:

Scenario	<i>Rug pull</i>	<i>Crabbing</i>	<i>To the moon</i>
Return r	−40%	+5%	+60%
$P(R = r)$	0.50	0.30	0.20

- Verify that P is a valid probability distribution.
- Compute $E[R]$. Based solely on expected return, is this a good “investment”?
- Compute $\text{Var}(R)$ and $\text{SD}(R)$. What does the SD tell you about this asset?
- The student invests \$500. Find the expected value and SD of their portfolio after one month.

Problem 3 Binomial Distribution – The Sleepy Barista [6 pts]

The campus coffee shop barista makes an incorrect order with probability $p = 0.12$, independently. Let X = number of wrong orders in the next $n = 10$; $X \sim \text{Binomial}(10, 0.12)$.

- Find $P(X = 0)$, the probability all 10 orders are correct. (4 decimal places.)
- Find $P(X \geq 2)$. (*Hint: complement rule.*)
- Find $E[X]$ and $\text{Var}(X)$. Interpret $E[X]$ in plain language.

- (d) The manager intervenes if *more than 2* orders are wrong. Find the probability of intervention.

Daily stream counts for mid-tier Ukrainian pop songs follow $N(\mu = 12,000, \sigma = 3,200)$.

- (a) What fraction of songs get **more than** 17,000 streams in a day?
- (b) Find the stream count at the **90th percentile**. (Use $z_{0.90} \approx 1.28$.)
- (c) Find $P(8,000 < X < 15,200)$.

Late-night Bolt Food delivery times follow $\text{Exponential}(\lambda)$ with mean $1/\lambda = 20$ minutes.

- (a) State $E[T]$ and $\text{Var}(T)$.
- (b) Find $P(T > 30)$: the probability of waiting more than 30 minutes.
- (c) Find $P(10 < T < 40)$: the probability of delivery in an “acceptable window.”
- (d) The student has already waited 20 minutes. Using the *memoryless property*, find $P(T > 30 \mid T > 20)$. Does the 20-minute wait change anything? Explain.

A Bolt Food courier tracks daily tip earnings T (UAH). Tips are **non-negative**. Historical records give **only**: $E[T] = 180$ UAH. The full distribution is **unknown**.

- (a) State Markov’s inequality (conditions and formula). Apply it: find an upper bound for $P(T \geq 900)$.
- (b) The courier’s friend claims: “*There is at most a 15% chance you earn more than 1,200 UAH in a day.*” Use Markov to check. Is the claim valid, too strong, or too weak?
- (c) Describe one specific distribution for T under which $P(T \geq 900)$ equals the Markov bound exactly. What does this illustrate about when Markov’s inequality is tight?

During exam season, KSE students sleep $\mu = 5.8$ h/night on average, $\sigma = 1.2$ h (distribution unknown). A researcher surveys $n = 36$ students.

- (a) State the approximate distribution of $\bar{X}$, name the theorem used, and give $E[\bar{X}]$ and $\text{Var}(\bar{X})$ numerically.
- (b) Find $P(\bar{X} < 5.5)$.
- (c) Find c such that $P(\bar{X} > c) = 0.05$. (Use $z_{0.95} \approx 1.645$.)

X = hours studying and Y = hours gaming on a random Saturday for KSE students.

$X \setminus Y$	$Y = 0$	$Y = 2$	$Y = 4$
$X = 1$	0.05	0.15	0.10
$X = 3$	0.10	0.25	0.15
$X = 6$	0.15	0.05	0.00

- (a) Find the marginal distributions of X and Y .
- (b) Compute $E[X]$ and $E[Y]$.
- (c) Are X and Y independent? Justify with one specific calculation.
- (d) Compute $\text{Cov}(X, Y)$ and interpret its sign.

$n = 25$ KSE students: X = cups of coffee/day, Y = self-rated productivity (0–100).

$$\bar{X} = 2.8, \quad \bar{Y} = 64.0, \quad S_X = 1.1, \quad S_Y = 12.5, \quad \sum_{i=1}^{25} (X_i - \bar{X})(Y_i - \bar{Y}) = +154.0.$$

- (a) Calculate the sample covariance S_{XY} .
- (b) Calculate the sample correlation r_{XY} .
- (c) Interpret the sign and magnitude of r_{XY} in context.
- (d) A student concludes: “More coffee causes higher productivity.” Identify the flaw.

A casino coin-flip game: each round you bet \$100. **Heads** ($p = 0.45$): net gain +\$100. **Tails** ($p = 0.55$): net gain −\$100. Let X_i be the net profit on flip i ; all flips are independent.

- (a) Find $E[X_i]$ and $\text{Var}(X_i)$. State in one sentence whose side the math is on.
- (b) A player makes $n = 100$ flips. Let $S_{100} = \sum_{i=1}^{100} X_i$. Compute $E[S_{100}]$ and $\text{Var}(S_{100})$, name the theorem that justifies the normal approximation, and state that distribution with all numerical parameters.
- (c) Find $P(S_{100} > 0)$. (*Hint: $\sqrt{990,000} \approx 995$.*)
- (d) Invoke the **Law of Large Numbers** to explain precisely what $\bar{X}_n$ converges to, and why the strategy “*keep playing until I win*” is mathematically doomed.

REFERENCE PAGES - Do Not Write Answers Here

Distribution	PMF / PDF	Mean	Variance
Bernoulli(p)	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	$(1-p)^{k-1}p, k = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$\frac{\lambda^k e^{-\lambda}}{k!}, k \geq 0$	λ	λ
Uniform(a, b)	$\frac{1}{b-a}, a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	μ	σ^2

Standard Normal Cumulative Distribution - $\Phi(z) = P(Z \leq z)$, $Z \sim N(0, 1)$

Area to the **left** of z . For negative z : $\Phi(-z) = 1 - \Phi(z)$. Example: $\Phi(-1.00) = 1 - 0.8413 = 0.1587$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Quantiles of the Standard Normal $z_p = \Phi^{-1}(p)$, i.e. $P(Z \leq z_p) = p$

p	0.90	0.95	0.975	0.99	0.995	0.999
z_p	1.282	1.645	1.960	2.326	2.576	3.090

Left tail: $\Phi^{-1}(1 - p) = -z_p$. *Example:* $\Phi^{-1}(0.05) = -1.645$.